

Week 1: Induction, Pigeonhole, and the Extremal Principle

Lecture notes · Thu Sep 24 / Fri Sep 25

Toolbox: three ways to prove something exists.

1. **Pigeonhole.** N objects in n boxes put at least $\lceil N/n \rceil$ in one box. Averaging form: if $y_1 + \cdots + y_n = s$, some $y_i \geq s/n$ and some $y_j \leq s/n$.
2. **The art is choosing the boxes:** residues mod n , odd parts, partial sums, fractional parts in short intervals, small cells of a region. Ask: *what do I want two objects to share?*
3. **Extremal principle.** A finite nonempty set of reals has a largest and smallest element; a nonempty set of positive integers has a least one. Pick the extreme object (longest, closest, largest area, least counterexample) and use that nothing beats it, usually by contradiction. “Sort the numbers” is this principle in disguise.
4. **Induction.** Ordinary, strong ($P(k)$ for all $k < n$ implies $P(n)$), or on a well-chosen parameter ($p + q$ for a fraction p/q , the largest element, the number of odd terms). It is also how “repeat this step” becomes a proof.
5. **Strengthen the hypothesis.** If the inductive step fails, prove something stronger: $\sum_{k \leq n} 1/k^2 < 2$ resists induction, but $\sum_{k \leq n} 1/k^2 \leq 2 - 1/n$ goes through in one line.

Example 1 (Pigeonhole with partial sums). Among any n integers $a_1, \dots, a_n$ some nonempty block of consecutive terms has sum divisible by n . Let $s_0 = 0$ and $s_k = a_1 + \cdots + a_k$. The $n + 1$ numbers $s_0, \dots, s_n$ fall into the n boxes “residue mod n ,” so two of them, s_i and s_j with $i < j$, are congruent mod n . Then $n \mid s_j - s_i = a_{i+1} + \cdots + a_j$, a nonempty consecutive block. Including $s_0 = 0$ is what makes the count $n + 1$.

Example 2 (Extremal principle in a tournament). In a round-robin tournament (every pair plays once, no ties) there is a player K such that every other player either lost to K or lost to someone who lost to K . Choose K with the maximum number of wins (possible: finitely many players). If some Y beat K and also beat everyone K beat, then Y has at least (wins of K) + 1 wins, contradicting maximality. So every Y lost to K or to somebody K beat. Nothing was constructed; we only used “no one has more wins than K .”

Example 3 (Strong induction, Putnam 2005 A1). Every positive integer is a sum of one or more numbers of the form $2^r 3^s$ ($r, s \geq 0$) in which no summand divides another. Strong induction on n ; $n = 1 = 2^0 3^0$. If n is even, double each summand of a representation of $n/2$; divisibility relations among summands are unchanged. If $n > 1$ is odd, let 3^s be the largest power of 3 with $3^s \leq n$. If $n = 3^s$, done. Otherwise $m = (n - 3^s)/2$ is a positive integer with $m < 3^s$ (as $n < 3^{s+1}$); take a representation of m , double it, and adjoin 3^s . A doubled summand $2^a 3^b$ has $a \geq 1$, so it does not divide 3^s ; and $3^b \leq m < 3^s$ gives $b < s$, so 3^s does not divide it. The doubled summands do not divide one another by induction. The step worked because we peeled off the largest power of 3, not because we looked at $n - 1$.

Pitfall. (1) Pigeonhole gives *two* objects in *one* box, not the two you wanted; make sure your conclusion follows for *any* two in a box (in Example 1, any $i < j$ works). (2) “Take the smallest counterexample” needs a well-ordered set: positive integers, yes; positive reals, no. (3) An extremal object exists only for finite sets or compact situations; say why it exists (“ S is finite, so some triple has maximal area”). (4) State $P(n)$ exactly and check the base case; the classic lost point is an inductive step that silently assumes $n \geq 2$.

How to recognize this technique on the exam. Pigeonhole is the first guess when a problem says “among any N of these, some two...” or hands you more objects than natural classes; look for a count one more than a modulus, a number of intervals, or a number of odd numbers. The extremal principle is the first guess when something must exist in a finite configuration (points, tournaments, colorings) and no formula is in sight: pick the largest, closest, or longest and argue it already works or could be improved. Induction is the natural shape when a statement indexed by n contains a smaller case inside it; if that sub-case is not “ $n - 1$,” expect strong induction or a nonstandard parameter.

Writing a solution that scores. State precisely what you will prove, including any reformulation (“We show that for all $n \geq 1$, ...”). Write full sentences with every quantifier visible; a page of equations with no words is not a proof. Justify every step a skeptical reader could question, naming the tool as you use it (“by pigeonhole on the $n + 1$ residues...”). Scores are 10 for a complete argument with cosmetic slips, 8–9 for a small gap, and almost always 0–2 otherwise: partial progress, an answer without proof, or the “easy half” rarely earns more than 1. Cross out what you abandon; never claim more than you have shown.

Suggested timing (15 minutes). 0–3: pigeonhole, Example 1, “choose the boxes.” 3–7: extremal principle, Example 2, sorting heuristic. 7–12: induction, Example 3, choosing the parameter and strengthening the hypothesis. 12–15: the writing box and exam format (Sat Dec 5, four 90-minute sessions with breaks, 12 problems at 10 points, median near 0); then groups start on Problems 1–4.